ALICE L WALTON SCHOOL OF MEDICINE

ECG Learning: Sinus Bradycardia and Tachycardia

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Learning Objectives (4)

After completing this brick, you will be able to:

1. Identify key ECG findings indicative of sinus bradycardia and tachycardia.
2. Explain how sinus bradycardia and tachycardia alters the ECG waveform and heart rate.
3. Describe the directionality of the ventricular depolarization vector and relation to the QRS axis.
4. Apply ECG interpretation skills to recognize sinus bradycardia and tachycardia on a 12-lead ECG.

CASE CONNECTION

JS, a 22-year-old male, presents to the university clinic with episodes of lightheadedness and fatigue during morning exercise. His symptoms are most pronounced after waking up early for his 6 AM training sessions. He has no significant past medical history and takes no medications. His vital signs show a blood pressure of 110/70 mmHg and a heart rate of 48 beats per minute.

Lee, a first-year medical student, encounters JS during a cardiology skills workshop and is tasked with interpreting his ECG results. What features on a 12-lead ECG help you identify sinus bradycardia? How would you recognize normal versus abnormal rhythm in this scenario? Consider your answers as you read, and we'll revisit JS at the end of the brick.

GO TO CONCLUSION ↓

Review:What are the limb and precordial leads and their locations?

A normal sinus rhythm ECG represents the electrical activity of a healthy heart. In sinus rhythm, the cardiac impulse originates in the sinoatrial (SA) node and follows the normal conduction pathway through the atria, atrioventricular (AV) node, bundle of His, bundle branches, and Purkinje fibers to the ventricles. This organized electrical activity creates a characteristic pattern of waveforms and intervals on the ECG that serve as the baseline for identifying abnormal cardiac rhythms.

ACTIVE LEARNING QUESTION

What is the correct sequence of waveforms in a normal sinus rhythm ECG?

- A. ○ T wave, P wave, QRS complex
- B. ○ QRS complex, P wave, T wave
- C. ● P wave, QRS complex, T wave
- D. ○ P wave, QRS complex, T wave, U wave
- E. ○ P wave, T wave, QRS complex

Explanation

The correct answer is P wave, QRS complex, T wave (C). This sequence represents the normal cardiac electrical activation pattern: atrial depolarization (P wave), followed by ventricular depolarization (QRS complex), and ventricular repolarization (T wave). Option (A) is incorrect because the T wave follows the QRS complex, not precedes it. Option (B) incorrectly places the P wave after the QRS complex. Option (D) incorrectly includes the U wave as a consistent component, while it's often absent or subtle. Option (E) incorrectly suggests the T wave precedes the QRS complex.

What Are Key ECG Findings Indicative of Sinus Bradycardia? (LO#1)

Sinus bradycardia is characterized by a normal sinus rhythm **with a heart rate below 60 beats per minute**. The primary defining feature is the slow heart rate, while the rhythm itself maintains the normal sequence of cardiac activation originating from the sinoatrial node. This distinction is crucial because it differentiates sinus bradycardia from other bradycardic rhythms that may have different origins or conduction patterns.

Bradycardia Diagnosing Features

In sinus bradycardia, each QRS complex is preceded by a P wave with normal morphology—typically upright in leads I and II and inverted in aVR. The PR interval remains within the normal range (120–200 ms) and is consistent throughout the tracin[illegible] from the atria to the ventricles proceeds normally, just at a slower rate.

The P waves themselves have a consistent morphology and each sinus P wave conducts to the ventricles with a resultant QRS complex, confirming that each atrial depolarization is followed by ventricular depolarization through the normal conduction pathway. Another way of describing this is that there is a consistent 1:1 relationship between the P wave and QRS complex (Figure 1).

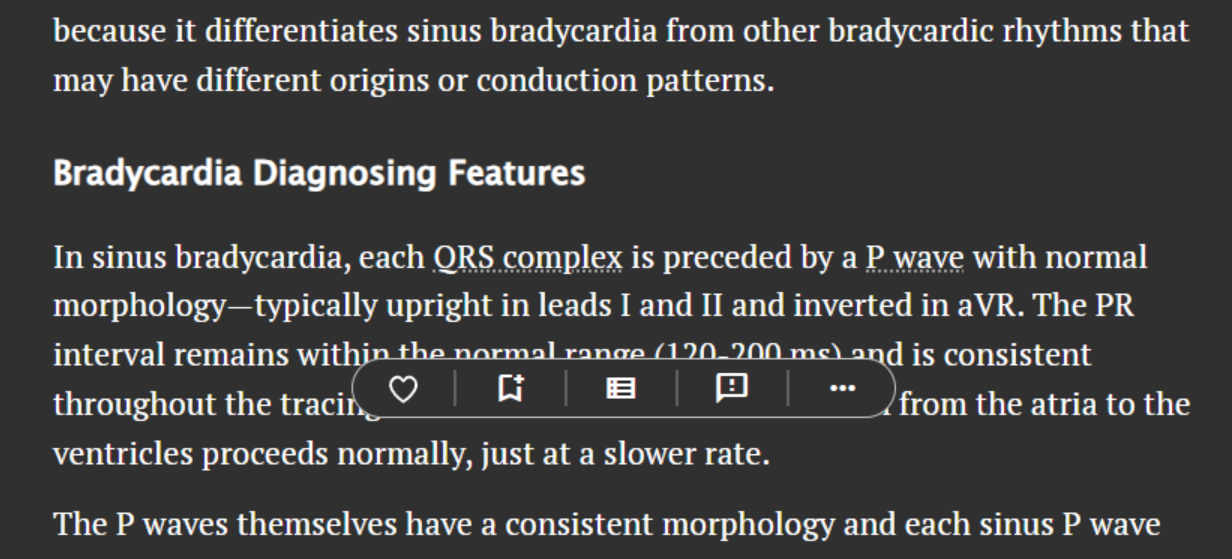


Figure 1: Sinus bradycardia with a 1:1 relationship between each P wave and QRS complex

Another key feature is that the P wave axis remains normal (0 to +90 degrees), indicating that the impulse originates in the SA node and follows the normal conduction pathway through the atria. **This distinguishes sinus bradycardia from junctional or ectopic atrial rhythms, where the P wave morphology or axis may be abnormal.** The QRS complexes in sinus bradycardia typically maintain normal duration (<120 ms) and morphology, indicating that ventricular depolarization proceeds through the normal conduction system despite the slower heart rate.

What is the P wave axis range in sinus bradycardia?

How Does Sinus Bradycardia Alter the ECG Waveform and Heart Rate? (LO#2)

Heart Rate Calculation

Heart rate calculation from an ECG can be performed using several methods. The most precise method involves measuring the R-R interval in seconds and dividing 60 by this value (heart rate = 60/R-R interval in seconds). For example, if the R-R interval measures 0.78 seconds (19.5 small boxes), the heart rate would be 77 beats per minute (60/0.78 = 77 bpm).

Figure 1. Alternatively, on a standard ECG paper (where each small box represents 0.04 secon[illegible]xes between consecutive R waves and dividing into 1500 yields the heart rate.

Figure 2. For quick estimation, counting the number of QRS complexes in a standard 10-second rhythm strip and multiplying by 6 provides the average heart rate in beats per minute. This method is useful when the R-R interval is irregular such as in arrhythmias like atrial fibrillation Figure 3.

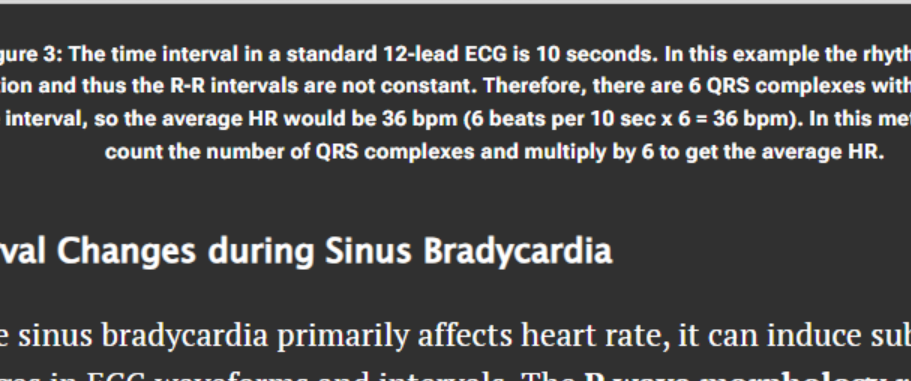


Figure 2: R-R method using small boxes

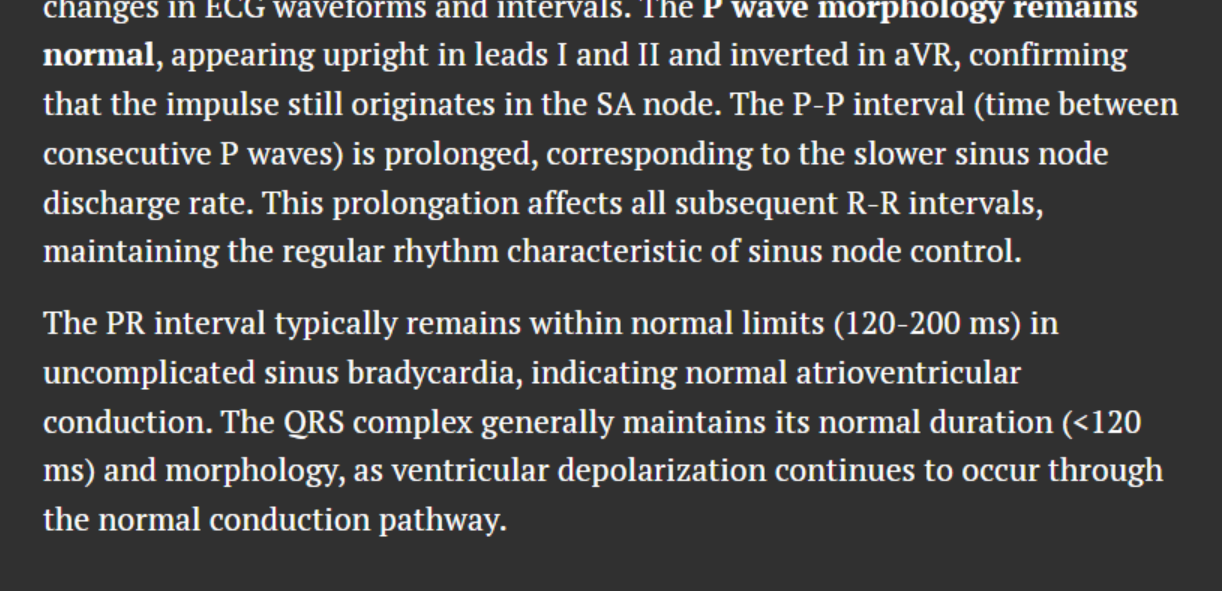


Figure 3: The time interval in a standard 12-lead ECG is 10 seconds. In this example the rhythm is atrial fibrillation and thus the R-R intervals are not constant. Therefore, there are 6 QRS complexes within a 10 second time interval, so the average HR would be 36 bpm (6 beats per 10 sec x 6 = 36 bpm). In this method, simply count the number of QRS complexes and multiply by 6 to get the average HR.

Interval Changes during Sinus Bradycardia

While sinus bradycardia primarily affects heart rate, it can induce subtle changes in ECG waveforms and intervals. The **P wave morphology remains normal**, appearing upright in leads I and II and inverted in aVR, confirming that the impulse still originates in the SA node. The P-P interval (time between consecutive P waves) is prolonged, corresponding to the slower sinus node discharge rate. This prolongation affects all subsequent R-R intervals, maintaining the regular rhythm characteristic of sinus node control.

The PR interval typically remains within normal limits (120-200 ms) in uncomplicated sinus bradycardia, indicating normal atrioventricular conduction. The QRS complex generally maintains its normal duration (<120 ms) and morphology, as ventricular depolarization continues to occur through the normal conduction pathway.

Stepwise ECG Analysis for Sinus bradycardia

A systematic approach to ECG interpretation for sinus bradycardia begins with heart rate assessmen[illegible]f the standard methods, such as the 1500 divided by R-R interval method or counting QRS complexes in a 10-second standard ECG strip and multiplying by 6. Confirm that the rate is below 60 beats per minute, meeting the primary criterion for bradycardia. For borderline cases (around 60 beats per minute), calculate the average rate over several consecutive cycles to ensure accuracy.

Step 1: Assess heart rate by measuring the R-R interval — use either the **1500 ÷ number of small boxes** between R-R waves or **count QRS complexes in a 10-second strip × 6** to estimate beats per minute.

Step 2: Confirm bradycardia when the calculated rate is **<60 bpm**, which meets the primary diagnostic criterion for sinus bradycardia.

Step 3: For borderline rates (~60 bpm), average several consecutive cycles to ensure accuracy and confirm consistent sinus rhythm.

Step 4: Evaluate rhythm regularity by inspecting **consecutive R-R intervals**—they should be **consistent** across the tracing in true sinus bradycardia.

Step 5: Allow minor variability (≤10% difference between shortest and longest R-R intervals), which can occur with **normal respiratory sinus arrhythmia**, especially in younger or healthy individuals.

Step 6: Identify irregularity—markedly varying R-R intervals may indicate an **alternative rhythm**, such as **atrial fibrillation with slow ventricular response** or **sinus arrhythmia with bradycardia**.

Step 7: Inspect P waves—ensure **each QRS complex is preceded by a P wave** with normal morphology (upright in leads I, II, aVF, V4–V6; inverted in aVR), reflecting SA node origin.

Step 8: Confirm normal P wave axis (0° to +90°) indicating **normal atrial depolarization** consi[illegible]

Step 9: Verify 1:1 conduction—P waves should have **consistent shape** and maintain a **1:1 relationship with QRS complexes**, confirming intact **SA-to-ventricular conduction** through the normal pathway.

What is the directionality of P wave and QRS Axis? (LO#3)

As mentioned earlier, impulses that are generated from the sinus node originate from a small cluster of specialized cells in the high right atrium. The sinus node, or sinoatrial node is normally located in the high lateral right atrium along the junction of the superior vena cava (SVC) and right atrium. After generation of this impulse, the depolarization wave front traverses through the atrial myocardium in an inferior and leftward direction. The word axis is used to describe the overall directionality of the depolarizing wave front, or vector, as it travels through myocardial tissue.

When the atria are depolarized by the sinus node, the overall direction of the traveling wave front is oriented down and leftward which manifest on the 12-lead ECG as positive P wave deflection in the inferior leads II, III and aVF as this wavefront is traveling towards these leads. In contrast, leads aVR and aVL have negative P wave deflections as the wave front is traversing away from these leads. In addition, since the overall propagating wavefront is also leftward, lead I will have a positive P wave as well. The overall normal axis of a P wave generated by the sinus node is between 0 - 90 degrees which will generate a positive P wave deflection in leads II and lead I **(Figure 4).**

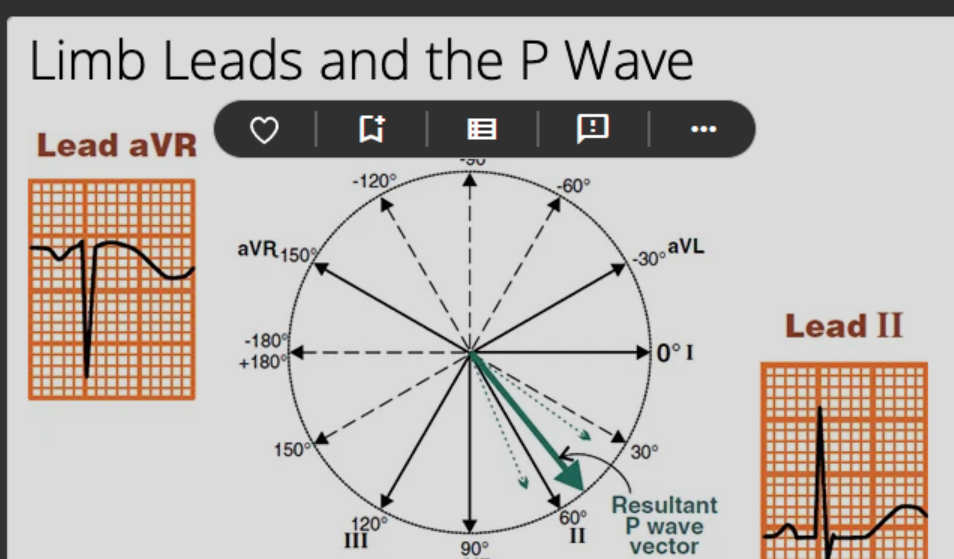


Figure 4: Overall P wave vector generated by the sinus node is directed inferior and leftward.

Just as a sinus P wave has a normal axis, a normal QRS complex has a normal axis which again, represents the overall directionality of the wave front propagation, but this time through ventricular myocardium. A normal QRS axis is between -30 degrees to +90 degrees, with the majority of time occurring between 0-90 degrees, similar to a normal P wave axis. This is manifested on the frontal leads as a predominately positive QRS wave in leads I and aVF or II because the overall depolarization wave front is traveling through the ventricular myocardium towards these leads, resulting in a mostly positive QRS deflection **(Figure 5)**. In contrast, leads aVR and aVL will have a predominantly negative QRS wave as the vector is traveling away from these leads.

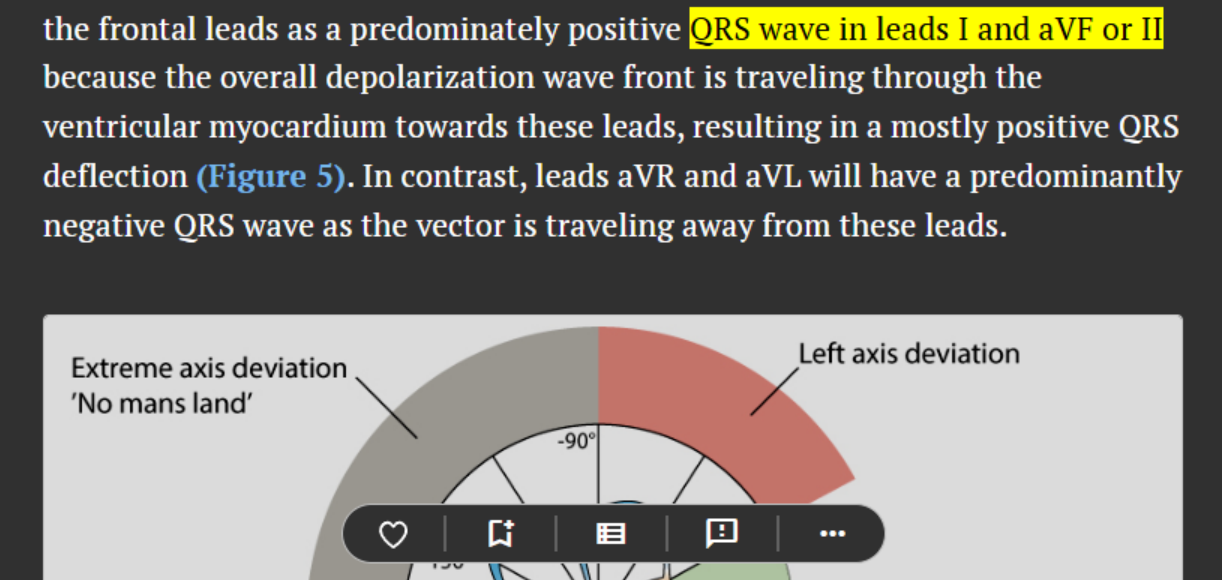

Extreme axis deviation 'No mans land'
Left axis deviation
-90°
180°
0°
I
30°
-aVR
60°
II
90°
aVF
120°
III
Right axis deviation
Normal electrical axis ranges between -30° to 90° (green area).

Figure 5: Normal QRS axis lays between -30 degrees and 90 degrees.

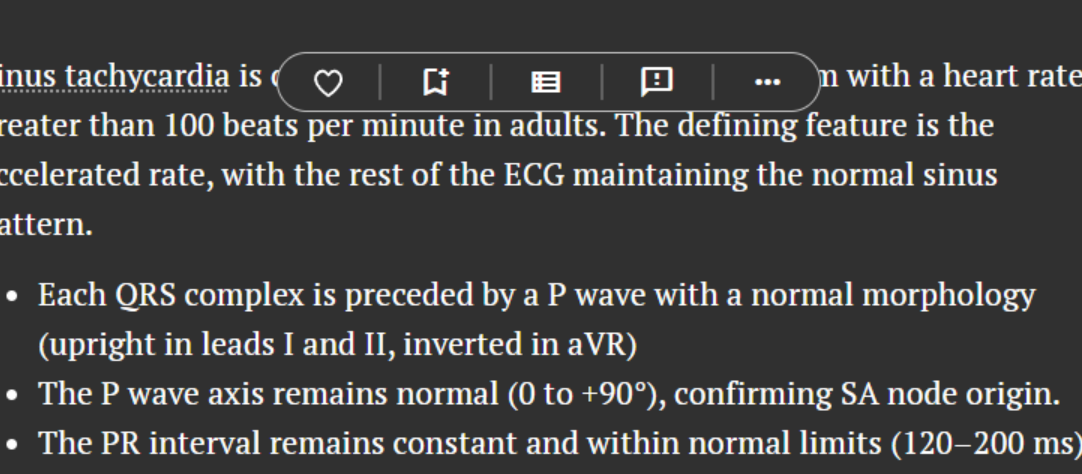
What are the key steps to confirm sinus bradycardia on ECG?

What Are Key ECG Findings Indicative of Sinus Tachycardia? (LO#1-3)

Sinus tachycardia is [illegible]n with a heart rate greater than 100 beats per minute in adults. The defining feature is the accelerated rate, with the rest of the ECG maintaining the normal sinus pattern.

- Each QRS complex is preceded by a P wave with a normal morphology (upright in leads I and II, inverted in aVR)
- The P wave axis remains normal (0 to +90°), confirming SA node origin.
- The PR interval remains constant and within normal limits (120–200 ms)
- There is a consistent 1:1 relationship between P waves and QRS complexes, demonstrating that every atrial depolarization is conducted to the ventricles.
- The QRS complex maintains normal duration (<120 ms) and morphology, reflecting normal ventricular conduction.

In essence, sinus tachycardia is a normal sinus rhythm at a faster rate. The rhythm remains regular, meaning the R-R intervals are consistent, but the interval between QRS complexes is shortened due to increased sinus node discharge rate.


Sinus Tachycardia
Rhythm: Regular
Rate: 130 bpm (normal range= 60-100 bpm)
P Waves: Upright and regular
P-R Interval: 0.16 sec (normal range= 0.12-0.20), one P wave for each QRS
QRS: 0.04 sec (normal range= 0.04-0.12)

Interval Chang[illegible]ia (LO#4)

Sinus tachycardia primarily affects the timing of cardiac depolarization and repolarization without altering the morphology of the waves.

- P-P Interval: The interval between consecutive P waves is shortened, reflecting increased sinus node firing.
- PR Interval: Usually remains within the normal range (120–200 ms), though at very high rates, it may shorten slightly due to faster AV nodal conduction.
- QRS Duration: Typically remains normal (<120 ms), since ventricular conduction proceeds normally
- QT Interval: The absolute QT interval shortens as the heart rate increases because ventricular repolarization occurs more rapidly. However, with the corrected QT (QTc), it generally remains within normal limits.

Applying ECG Interpretation Skills to Recognize Sinus Bradycardia and Tachycardia (LO#1, 2, and 4)

Recognition of sinus tachycardia—and distinction from sinus bradycardia—requires systematic analysis. The same stepwise approach can be used for all sinus rhythms, adjusting focus on rate determination.

- **Rhythm:** Confirm presence of a P wave before each QRS complex. Ensure P wave morphology and axis are consistent with SA node origin (upright in I and II, inverted in aVR).
- **Rate:** Determine heart rate using one of the methods above. Confirm rate >100 bpm for sinus tachycardia. For sinus bradycardia, confirm rate <60 bpm.
- **P:QRS Relations**[illegible] P waves and QRS complexes. Ensure each sinus impulse is conducted to the ventricles.
- **Regularity:** Check R-R intervals for regularity. In sinus tachycardia, the rhythm remains regular, though mild variability may occur with respiration or autonomic tone changes.
- **Interval Measurements:** PR interval: 120–200 ms (may shorten slightly with sinus tachycardia). QRS duration: <120 ms. QT interval: Shortens proportionally with faster heart rate (QTc usually normal).

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to JS, what features on a 12-lead ECG help you identify sinus bradycardia? How would you recognize normal versus abnormal rhythm in this scenario? Using systematic ECG analysis, note JS's heart rate below 60 beats per minute with regular P waves preceding each QRS complex. These findings confirm sinus bradycardia and guide your clinical assessment.

The ECG shows a regular rhythm at 48 beats per minute with normal P wave morphology, consistent PR interval (160 ms), and normal QRS complexes. Given JS's age, athletic status, and absence of concerning symptoms beyond mild lightheadedness during morning exercise, this likely represents physiological sinus bradycardia due to increased vagal tone and athletic conditioning rather than pathological bradycardia requiring intervention.

Summary

- Limb leads (I, II, III, aVR, aVL, aVF) and precordial leads (V1-V6) are placed on the chest wall[illegible]
- A normal sinus rhythm ECG shows the sequence P wave, QRS complex, T wave, representing atrial depolarization, ventricular depolarization, and ventricular repolarization.
- Sinus bradycardia is characterized by a heart rate below 60 bpm, maintaining normal P wave morphology and consistent PR intervals.
- In sinus bradycardia, the ECG shows a regular rhythm with prolonged R-R and P-P intervals, while the QT interval may increase in absolute duration.
- Systematic ECG interpretation for sinus bradycardia includes assessing heart rate, rhythm regularity, P wave morphology, and conduction intervals.
- Common pitfalls in diagnosing sinus bradycardia include mistaking it for junctional rhythm, overlooking AV blocks, and misinterpreting artifacts.

Review Questions

1. What ECG finding best indicates sinus bradycardia?

- A. ● Heart rate below 60 bpm
- B. ○ QRS duration over 120 ms
- C. ○ Irregular rhythm pattern
- D. ○ Absent P waves before QRS
- E. ○ Inverted P waves in lead II

Hide Explanation

The correct answer is (A) Heart rate below 60 bpm with normal P waves. Sinus bradycardia has a slow heart rate (<60 bpm) with normal P wave morphology preceding each QRS complex. (B) QRS duration >120 ms indicates conduction delay, not sinus bradycardia. (C) Irregular rhythm excludes sinus bradycardia. (D) Absence of P waves suggests junctional or atrial fibrillation. (E) Inverted P waves indicate ectopic rhythm.

2. What is a key step to confirm sinus bradycardia on a 12-lead ECG?

- A. ○ Heart rate less than 60 bpm
- B. ● Normal P waves before QRS
- C. ○ Consistent regular rhythm
- D. ○ Normal QRS morphology
- E. ○ Prolonged QT interval

Hide Explanation

The correct answer is (B) Identify normal P waves before each QRS. This confirms sinus origin of the rhythm. (A) Heart rate <60 bpm is necessary but not sufficient alone. (C) Regular rhythm supports sinus bradycardia but must be combined with normal P waves. (D) Normal QRS morphology supports normal conduction but doesn't confirm sinus[illegible] but not diagnostic.

3. What is a typical P wave axis in sinus bradycardia?

- A. ○ Less than 0 degrees
- B. ● 0 to +90 degrees
- C. ○ Greater than +90 degrees
- D. ○ -30 to +30 degrees
- E. ○ Less than -90 degrees

Hide Explanation

The correct answer is between 0 to +90 degrees (B). This axis supports sinus origin of the rhythm. Less than 0 degrees (A) or greater than +90 degrees (C) suggests an ectopic origin. Between -30 to +30 degrees (D) is too narrow and unlikely for normal P waves. Less than -90 degrees (E) indicates abnormal depolarization direction.

4. Which interval shortens in sinus tachycardia?

- A. ● P-P interval
- B. ○ QTc interval
- C. ○ PR interval
- D. ○ QRS duration

Hide Explanation

The correct answer is P-P interval (A). The P-P interval (A) shortens reflecting increased sinus node firing. In sinus tachycardia, the QT interval shortens, however QTc correction the limits are within normal range. The PR interval (B) usually remains normal, and the QRS duration (D) typically remains normal.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 ☑ ~~Identify key ECG findings indicative of sinus bradycardia and tachycardia.~~
- 02 ☑ ~~Explain how sinus bradycardia and tachycardia alters the ECG waveform and heart rate.~~
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- 03 ☑ ~~Describe the directionality of the ventricular depolarization vector and relation to the QRS axis.~~
- 04 ☑ ~~Apply ECG interpretation skills to recognize sinus bradycardia and tachycardia on a 12-lead ECG.~~

Objective Confidence: 4/4